METRIC

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MILITARY SPECIFICATION

BATTERY CELLS, STORAGE, SILVER-ZINC ALKALINE TYPE (FOR NR-1) (METRIC)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers silver-zinc alkaline storage battery cells for use in the assembly of the NR-1 submarine storage battery.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-380 - Plastic Molding Material Methacrylate.
QQ-B-613 - Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet and Strip).
QQ-B-626 - Brass, Leaded and Nonleaded: Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar and Strip).
QQ-B-637 - Brass, Naval: Rod, Wire, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire, and Strip).
QQ-B-639 - Brass, Naval: Flat Products (Plate, Bar, Sheet, and Strip).
QQ-B-650 - Brazing Alloy, Copper, Copper-Zinc, and Copper-Phosphorus.
QQ-B-654 - Brazing Alloys, Silver.
QQ-B-825 - Bus Bar: Copper, Aluminum, or Aluminum-Alloy.
QQ-C-576 - Copper Flat Products With Slit, Slit and Edge-Rolled, Sheared, Sawed, or Machined Edges (Plate, Bar, Sheet and Strip).
QQ-N-290 - Nickel Plating (Electrodeposited).
QQ-S-365 - Silver Plating, Electrodeposited, General Requirements For.
PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-300 - Chemicals, Liquid; Packaging and Packing of.
PPP-C-650 - Crates, Wood, Open and Covered.

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MIL-P-116 - Preservation-Packaging, Methods of.
MIL-C-3774 - Crates, Wood, Open 12,000- and 16,000-Pound Capacity.
MIL-B-7883 - Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-C-15816 - Covers, Complete, and Parts, Hard Rubber for Use with Submarine Storage Batteries.
MIL-P-15817 - Parts, Synthetic Rubber, for Use with Batteries or in Battery Compartments of Submarines (METRIC).
MIL-M-24041 - Molding and Potting Compound, Chemically Cured, Polyurethane.
MIL-P-46120 - Plastic Molding and Extrusion Material, Polysulfone.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6140

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STANDARDS
MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORTATION (DOT)
Code of Federal Regulations-Title 49
Section 173.260, subpart E

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

NATIONAL CLASSIFICATION BOARD
National Motor Freight Classification Classes and Rules

(Application for copies should be addressed to the ATA Tariff Section, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, G. F. Earl, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production 6 cells shall be examined and tested as specified in 4.3 (see 6.3).

3.2 General requirements.

3.2.1 Design and construction. The battery installation shall consist of 150 separate series-connected cells with a capacity of 850 ampere hours (Ah) at the 5-hour rate. Terminal parts (nuts, bolts, and washers) shall be furnished. The dimensions, the location and spacing of the terminals extending above the cell cover shall be as shown on figure 1.

3.2.2 Installation and interface. After delivery, the cells will be connected in the ship's battery compartment by the installing activity. The battery will always be connected to the busbar, whether on charge or discharge.

3.3 Electrical requirements.

3.3.1 Electrical characteristics. The discharge characteristics of the 150-cell battery, based on an initial electrolyte temperature and an ambient temperature of 25°C ± 5°C at the start of the discharge, shall be at least that shown in table I.

TABLE I. Discharge-characteristics.

Discharge		Capacity (Ah)	Battery average voltage (V)	Minimum average[1/] cell voltage (V)	Minimum (cut-off) cell voltage (V)
rate amperes (A)	Time hours (h)				
275	3	825	210	1.42	1.25
170	5	850	210	1.45	1.31
87.5	10	875	213	1.48	1.36
45	20	900	213	1.50	1.39

See footnote on next page.

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1/ The "average voltage" is defined as the average of voltage recordings taken at equal intervals, starting with the first recording (taken between 1/2 minute and 1 minute after applying a constant rate) and including the last recording. The interval of recording shall be not greater than h/5 hours where h is the time in hours of the rated capacity test. The time to the last recording shall be not greater than the rated time.

3.3.2 Life. The battery, when installed, shall be capable of delivering a cumulative total of 50,000 Ah of discharge over a period of 24 months. Except as required by 4.6.3.2, the capacity shall be at least 90 percent of the rated capacity shown in table I for the first 25,000 Ah or 12 months of service, whichever occurs first. Thereafter, the capacity shall be at least 80 percent until the end of the guaranteed life (see 6.4).

3.3.3 Charge.

3.3.3.1 Full charge definition. Full charge is defined as an Ah input at least equal to 90 percent of the Ah output obtained in the prior discharge.

3.3.3.2 Charge acceptance. After a discharge at any of the rates to the corresponding cut-off voltages shown in table I, and using any charging rates as selected by the Naval Ship Engineering Center (NAVSEC) up to 60 A, the battery or single cells shall be capable, starting at an ambient temperature of 25°C ± 5°C, of accepting full charge as defined in 3.3.3.1 in 18 hours or less.

3.3.4 Cell terminals. The current density of any individual terminal shall be such that the temperature at the top of the positive terminal does not exceed 60°C at any time during a discharge at 275 A when tested as specified in 4.6.3.7.

3.3.5 Hydrogen evolution. Hydrogen evolution from any charged cell shall not exceed 50 milliliters (ml) per hour average over a 24-hour period when the cell is maintained on open circuit at 25°C ± 3°C.

3.3.6 Metal insulation. Terminals (noncontact surfaces) and other metal hardware exterior to the jar and cover shall be insulated with a nonconducting material to prevent contact with potassium hydroxide (KOH).

3.3.7 Internal cell resistance. The internal cell resistance of all cells shall not deviate more than 20 percent of the average value when measured as specified in 4.6.4.2.

3.3.8 Temperature monitoring. For the battery installation of 150 series-connected cells, 6 cells shall be used for temperature monitoring. All terminal studs of each of the 6 cells shall be in accordance with figure 2. The monitoring system that connects to these 6 cells will be supplied by the installing activity.

3.4 Physical requirements.

3.4.1 Dimensions. The cell dimensions shall be as shown on figure 1.

3.4.2 Mass. Any cell, when filled, shall not exceed 14.6 kilograms (kg), while the average mass of all cells, in the same condition, shall not exceed 14.3 kg.

3.4.3 Cell jar. Cell jar shall be molded out of polysulfone in accordance with MIL-P-46120, using the Government-owned mold (see 3.8). Typical configuration is shown on figure 3.

3.4.4 Cell cover. Cell covers shall be molded out of the same material as the cell jar using the Government-owned mold (see 3.8). Typical configuration is shown on figure 4.

3.4.5 Cover-to-jar sealing. The cover-to-jar sealing shall be epoxy or cement.

3.4.6 Electrode pack.

3.4.6.1 Grids. Silver shall be used for the positive and negative grids supporting the active materials. The equivalent thickness of each grid (positive or negative), when reduced to a solid sheet of same mass and same apparent electrode area, shall be at least 0.05 millimeters (mm).

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3.4.6.2 Tabs. Silver shall be used for the positive and negative electrode tabs attached to the terminal base. The tab area below the top of the grid shall be at least 90 percent hot-forge bonded to the grid.

3.4.6.3 Silver active material. Silver active material amount shall be at least 3.1 grams per Ah, based on the 20-hour rate capacity.

3.4.6.4 Electrode construction. The positive electrodes shall have smooth edges and corners rounded to a radius of at least 6 mm. The negative electrode grids shall have smooth edges and corners rounded to a radius of at least 6 mm.

3.4.6.5 Separator system. The separator system shall be as follows:

(a) The separator system shall consist, from positive electrode to negative electrode, of a positive absorbent material layer, cellophane-type main separator layers and a negative absorbent material layer.
(b) A layer of fuel cell grade asbestos, of 0.25 mm-nominal thickness, from Johns-Manville Corporation (with a maximum iron content of 0.5 percent by mass), or equal, shall be placed before or after the positive absorbent material specified in (c). The top shall be 5 to 6 mm over the top of the electrodes.
(c) The positive and negative absorbent material shall be 0.12 to 0.17 mm-thick. The material shall be a layer of non-woven polyamide material from Pellon Corporation, or equal. The top shall be 5 to 6 mm over the top of the electrodes. The material shall be heat-sealed along the bottom and the two vertical edges to form a bag around each electrode.
(d) The main separator shall be Yardney Electric Corporation C-19, Union Carbide Fibrous Sausage Casing (battery grade), or equal. The overall wet thickness of the combined layers of this material when swollen in the specified electrolyte, shall be at least 0.5 mm. The main separator height over the top of the electrodes shall be at least 25 mm.
(e) The main separator assembly method shall consist of bagging or wrapping the positive and negative electrodes separately. If bagging and wrapping of a pair of electrodes is used, it shall be done on the positive electrodes. If wrapping is used, the overlap of the last layer shall be at least 15 mm.
(f) For separator top protection, the contractor shall use strips of 0.03 mm (maximum)-thick microporous polypropylene film material (Celgard 3400 from Celanese Plastic Corporation or RAI-2291 from Radiation Research Corporation, or equal) of an electrical resistivity not higher than 130 ohm-centimeters in the specified electrolyte. One 90 mm ± 1 mm wide strip shall be inserted flush with the top edge of the main separator between two consecutive layers of the main separator.
(g) Epoxy or cement sealing of the bottoms of separators shall not be used.

3.4.7 Terminals. Current carrying terminal parts shall be made of copper or copper alloys in accordance with QQ-B-613, QQ-B-626, QQ-B-637, QQ-B-639, QQ-B-825, or QQ-C-576. Terminal parts shall be silver-plated in accordance with QQ-S-365, type III, grade B, with a nickel undercoat in accordance with QQ-N-290, class 1, grade F, form SB. Terminals shall be rectangular in shape to extend above the jar wall by 8 mm nominal. The volume of the well on the cover top around each terminal and within and up to the height of the retaining walls shall be filled with polyurethane potting material in accordance with MIL-M-24041. Non-contact terminal surfaces above the potting material shall be insulated in accordance with 3.3.6. Terminal dimensions and location shall be in accordance with figure 1. Brazing of terminal parts shall be in accordance with MIL-B-7883, grade B, using brazing alloys in accordance with QQ-B-650 or QQ-B-654. The terminals shall have threaded studs 21.5 mm ± 0.5 mm above the flat terminal contact surface. Plate tabs shall be attached to the base of a single solid copper strap for each polarity. Soldering shall not be used for attachment of plate tubes. Positive and negative terminals shall be identified.

3.4.8 Rubber parts. Soft rubber parts shall conform to MIL-P-15817. Hard rubber parts shall conform to MIL-C-15816. The O-rings shall be made of ethylene-propylene.

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3.4.9 Electrolyte.

3.4.9.1 Composition. The electrolyte used for filling the cells shall be an aqueous KOH solution of 45 percent ± 1 percent KOH by mass with no additives. The maximum concentration of impurities allowable in the electrolyte prior to filling shall be as listed in table II.

TABLE II. Maximum concentration of impurities in electrolyte.

Impurities	Maximum concentration
Potassium carbonate	0.50 percent
Chloride (as Cl)	0.005 percent
Sulfate (as SO_4)	0.002 percent
Nitrogen compounds (as N)	0.001 percent
Ammonium hydroxide precipitate	0.01 percent
Phosphate (as PO_4)	5 parts per million (p/m)
Iron	5 p/m
Sodium	0.02 percent
Other metals total combined	2 p/m
Reducing material	See note

Note: Reducing material shall be determined at 75°C ± 5°C in an acid medium by a maximum of 1.5 ml of 0.01 normal potassium permanganate solution reacting completely with reducing material contained in a 10 ml sample of KOH solution.

3.4.9.2 Level indicator. Each cell shall be fitted with an electrolyte level indicator molded of material in conformance with L-P-380, using the Government-owned mold (see 3.8). The level indicator shall visually indicate 4 electrolyte levels: too high, too low, and two intermediate positions. It shall be possible to install the level indicator in any rotational position for the best viewing angle when cells are installed in an inaccessible location of the battery tank. The level indicator shall be an integral part of the flash arrester assembly in accordance with figure 5 and shall be easily removable for cleaning The level indicator shall be in accordance with figure 6. Quantities of level indicators shall be as follows:

(a) Type A - 44.
(b) Type B - 106.

Provisions shall be made in each cell for use of one alternate means for determining the real electrolyte level if too low to be reached by the first level indicator or if the first level indicator is destroyed, damaged, or not usable. This alternate means may be independent of the cell and fitted to the cell when needed.

3.4.9.3 Level. The specified level and free amount of electrolyte in the cell shall be such that adequate self-irrigation will occur during normal operation of the cells (end of charge and high-rate discharge). At no time during operation shall the electrolyte level variation be more than 50 mm from the highest to lowest level.

3.4.9.4 Temperature. On charge or discharge, the maximum temperature measured in the electrolyte of any cell shall not exceed 70°C.

3.4.10 Cooling. No cell cooling device requiring the circulation of coolant is permitted.

3.4.11 Firewalls. Each cell shall be provided with internal firewalls between the cell jar and the cell pack to protect the bottom and the four vertical sides. The firewalls shall extend at least 6 mm above the separator top. The firewalls shall consist of three layers of 0.25 mm-nominal thickness, fuel cell grade asbestos from Johns-Manville Corporation, or equal, enclosed in a 0.1 mm-nominal thickness polyethylene sheet envelope, heat-sealed on all edges except the vertical firewalls perpendicular to the electrodes which shall have a plastic sheet or netting (0.25 mm-nominal thickness) between two layers of asbestos to give support and rigidity to the assembly. Pinholes are allowed at the top of all vertical firewalls to let air out for ease of assembly.

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3.4.12 Flash arresters. The cell cover shall be fitted with a flash arrester which shall be the only fitting which will permit the escape of gas from the cell during normal operation. The flash arrester shall have a vent plug base, in accordance with figure 7 molded out of a thermoplastic material resistant to the electrolyte, using the Government-owned mold (see 3.8).

3.4.12.1 Material. Flash arrester material shall be Cortland Grinding Wheel Company, nonwetting fused, aluminum oxide number A-46-V or Simonds Abrasive Company, blue bond type fused, non-wetting, aluminum oxide, grit size 46, or equal.

3.4.12.2 Pressure drop.

3.4.12.2.1 Dry flash arrester. A dry flash arrester at room temperature, shall not cause a pressure drop greater than 100 pascals (Pa) when tested as specified in 4.6.3.8.1.

3.4.12.2.2 Wet flash arrester. A wet flash arrester, at room temperature, shall not cause a pressure drop greater than 1000 Pa when tested as specified in 4.6.3.8.2.

3.4.12.3 Flash arresting ability. Flash arresters shall prevent ignition of gases on the underside of the arrester when tested as specified in 4.6.3.8.3.

3.4.13 Gas-tightness. The cells shall be capable of withstanding 35 kilopascals (kPa) differential air pressure for 5 minutes without loss of pressure.

3.4.14 Cell identification.

3.4.14.1 Cell markers. A set of individual cell markers shall be furnished and shall be consecutively numbered from 1 through 150. Cell markers shall not be attached to the cells.

3.4.14.2 Serial numbers. The manufacturer's serial numbers shall be visible from the top of the installed cells and shall be indelibly marked. The serial number shall consist of 5 digits: the first 2 digits represent the year of manufacture and the last 3 digits shall be serially assigned by the manufacturer.

3.4.15 Mercury.

3.4.15.1 Content. The amount of mercury or mercury substitutes shall be as specified (see 6.2.1) in a quantity indicated as a percent of the total mass of zinc oxide used per cell.

3.4.15.2 Labeling and marking. If mercury is used, each cell shall have a label, resistant to the electrolyte, and approximately 60 by 100 mm in size. Stencil may be used if resistant to the specified electrolyte. The format shall be as follows and shall include the name and amount of the mercury compound used in each cell:

CAUTION - MERCURY

CONTAINS MERCURY COMPOUND
IN NEGATIVE ELECTRODES

NAME: _____ AMOUNT_____grams

_____

USE AUTHORIZED BY NAVSEA

_____

IN THE EVENT OF A SHORT CIRCUIT RESULTING IN SMOKE OR FIRE, ISOLATE THE AREA AND NOTIFY THE COMMANDING OFFICER OR OFFICER-IN-CHARGE, MEDICAL OFFICER, AND QUALITY ASSURANCE DEPARTMENT.

3.5 Environmental requirements.

3.5.1 Temperature. The battery shall be capable of withstanding temperatures up to 70°C for 2 hours at any high rate discharge while the range for normal operation is 15°C to 50°C.

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3.5.2 Pressure. The battery shall operate in the submarine internal ambient pressure.

3.5.3 Angle of operation. The battery shall be capable of withstanding an angle of 45 degrees from the vertical in any direction without physical or electrical damage or electrolyte spillage from any cell.

3.6 Cells, repair parts, and accessories.

3.6.1 Cells. A total of 159 cells shall be delivered. The 159 cells shall be divided as follows:

(a) 150 cells making up a battery for installation in the ship. The cells shall be shipped as specified (see 6.2.1).
(b) 9 spare cells, for possible replacement of failed cells in the installed battery, shall be kept dry and shipped as specified (see 6.2.1).

3.6.2 Repair parts. Repair parts are those parts needed for repair, replacement, maintenance, and installation. The repair parts shall be shipped as specified (see 6.2.1) and include, as a minimum, the following:

(a) 30 flash arresters and O-rings.
(b) 150 electrolyte level indicators and O-rings (100 type B, 50 type A) (see figure 6).
(c) Requisite electrolyte for all dry cell filling (to be provided in separate bottles of 3 to 4 liters).
(d) 2 auxiliary level indicator assemblies.
(e) 9 cell marker blanks.
(f) 2 electrolyte adjustment devices.
(g) Materials or parts deemed necessary by the manufacturer for repair, replacement, maintenance and installation.

3.6.3 Accessories. The following accessories (hardware parts for 250 cells) shall be shipped as specified (see 6.2.1):

(a) Jar.
(b) Cover.
(c) Flash arrester.
(d) Electrolyte level indicators, (see figure 6): type A - 100 each, type B - 150 each.

3.7 Engineering services (see 6.2.1). Services of the contractor's engineer shall be available upon NAVSEC's request in order to:

(a) Assist the shipyard in preparing the battery cells for installation and installing them.
(b) Assist the ship's force in becoming familiar with the battery.
(c) Take corrective action in case of unexpected problems.
(d) Examine the battery condition when requested.
(e) Participate in technical meetings with NAVSEC or the ship's force with regard to the battery and in locations other than the manufacturer's plant.

A maximum of 25 mandays shall be allowed for the entire contract or life of the battery. Field engineering travel reports shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2).

3.8 Government-owned molds. The Government shall make available the following Government-owned molds:

(a) Cell jar (see 3.4.3).
(b) Cell cover (see 3.4.4).
(c) Electrolyte level indicator (see 3.4.9.2).
(d) Vent plug for flash arrester (see 3.4.12).

3.9 Technical data. The contractor shall prepare technical data in accordance with the data ordering documents included in the contract or order (see 6.2.2) and as specified in 3.9.1 through 3.9.4. The contractor shall use the International System of Units (SI) in all technical data. For threaded hardware designation, the Unified Thread Standard shall be used.

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3.9.1 Drawings. The contractor shall prepare engineering drawings (including document referenced therein) for review by NAVSEC in accordance with the data ordering document included in the contract or order (see 6.2.2) and as described in 3...1.1 and 3.9.1.2.

3.9.1.1 Nonproprietary documents. Nonproprietary documents shall not mention proprietary information or refer to any other document not supplied. Nonproprietary documents shall include, but not be limited to, the following:

(a) Cell jar.
(b) Cell cover.
(c) Level indicator.
(d) Flash arrester.
(e) Terminal.
(f) Sketch showing relative positions of tops of plates and separators with respect to cover underside.
(g) Terminal-cover assembly.
(h) Cell control drawing.
(i) Cell pack assembly.
(j) Firewalls.
(k) Electrode and tab dimensions.
(l) Characteristics of cell electrode pack components: number of electrodes; mass of active material per electrode; type, number and thickness of absorbent materials and separators between electrodes; grid type used in electrodes and its mass in grams per square centimeter g/cm^2.
(m) Data list (list of all drawings that are used in the manufacture of the cells).

3.9.1.2 Proprietary documents. Proprietary engineering drawings (including documents referenced therein) shall be accessible to NAVSEC for review at any time.

3.9.2 Service manual. The contractor shall prepare a service manual in accordance with the data ordering document included in the contract or order (see 6.2.2) and the appendix to this specification.

3.9.3 Milestone plan. The contractor shall provide and maintain a milestone plan and prepare a report in accordance with the data ordering documents included in the contract or order (see 6.2.2).

3.10 Design review. Design review meetings with Government representatives shall be held prior to the release of any drawings to vendors, or production, or to finalization of the service manual. At least 3 meetings (2 at NAVSEC) are required: one to establish a guideline agreement prior to design, one to review drawings and design, and one to discuss the service manual.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program. The contractor shall provide and maintain a quality program in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 First article inspection. Six cells for the entire contract shall be subjected to the first article inspection which shall consist of the examination specified in 4.6.2.1 and the tests specified in 4.6.2.2 and 4.6.3 through 4.6.3.8.3. Within 15 days after completion of the first article inspection, the cells shall be shipped with all necessary parts and instructions to a Government laboratory (see 4.7) for life testing (see 4.8).

4.4 Quality conformance inspection. The entire production lot shall be subjected to the inspection of 4.6.2. The sample cells (see 4.3.2) shall be filled, formed, and subjected to the inspections specified in 4.6.4. After completion of the quality conformance inspection, the sample cells shall be shipped to a Government laboratory (see 4.7) for life testing (see 4.8). When notified (see 4.6.4.4), the 150 cells shall be assembled as a battery and subjected to the tests specified in 4.6.4.

4.4.1 Sampling for quality conformance inspection. A sample of 6 cells shall be randomly selected from the production lot, representing each 150-cell battery plus the 9 spare cells, for quality conformance inspection (see 4.4) and life testing (see 4.8).

4.5 Battery acceptance.

4.5.1 Acceptance in the dry condition. Prior to acceptance by the cognizant Defense Contract Administration Service (DCAS), the technical data specified in 3.9.1, 3.9.2, 4.10, 4.11.1, 4.11.2, and 4.11.4 shall have been delivered and accepted by NAVSEC, the cells in the dry condition shall have passed the examination and tests specified in 4.6.2, and the sample cells shall have passed the tests specified in 4.6.4. After acceptance of the cells as a battery by the cognizant DCAS, the contractor shall be responsible for the storage of the dry cells in proper conditions and their subsequent physical and electrical performance after the cells are filled and formed and until accepted again in accordance with 4.5.2.

4.5.2 Acceptance in the wet condition. Upon written notification by the Naval Sea Systems Command (NAVSEA), within 1 year after acceptance in the dry condition, the contractor shall be responsible for filling and forming the cells as a battery as required by 4.6.4.1. After demonstration of performance in accordance with 4.6.4.1, 4.6.4.2 and 4.6.4.5, the cognizant DCAS will then accept the cells for delivery and shipment as directed by NAVSEA.

4.6 Inspection methods.

4.6.1 Inspection of parts. Parts used in the assembly of the cells shall be examined on a sample basis in accordance with MIL-STD-105, inspection level II acceptance quality level (AQL) equals 1.0 percent defective for conformance to the requirements of this specification and to the manufacturer's quality control requirements.

4.6.2 Inspection of all cells in the dry condition.

4.6.2.1 Examination. Cells (first article, sample, and production lot) shall be examined for workmanship and for conformance with 3.3.6, 3.4.1, and 3.4.14. Any defective cell shall be rejected.

4.6.2.2 Gas-tightness. Each cell shall be tested to determine conformance with 3.4.13.

4.6.3 Inspection of first article cells in the wet condition. Except where otherwise specified, the temperature of the electrolyte at the start of each test shall be within 16°C to 32°C.

4.6.3.1 Mass. The first article cells shall be weighed after formation. The mass shall meet the requirements specified in 3.4.2.

4.6.3.2 Capacity and voltage determination. After formation, in accordance with the manufacturer's instructions and the test specified in 4.6.4.1, each cell shall be recharged and then discharged at 275 A, recharged and then discharged at 170 A. The discharges shall be made to the corresponding cut-off voltages (see 3.3.1). Voltage, current, and electrolyte temperature shall be monitored and recorded. The results of the capacity discharge tests shall be averaged and the estimated standard deviation calculated according to the following formula:

$$s = \left(\frac{\sum_{1}^{n} (x_i - \bar{x})^2}{n-1} \right)^{1/2}$$

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Where:

s = Estimated standard deviation.
n = Number of sample cells.
$\bar{x}$ = Average of results of tests.
X_i = Result of individual sample cell.

The capacity, voltage, and electrolyte temperature of each cell shall meet the requirements of 3.3.1 and 3.4.9.4 and the average percent of capacity for all cells shall be equal to or greater than 100 percent of rated capacity plus 2.45 s.

4.6.3.3 Charge acceptance. Each cell shall be tested for charge acceptance to determine conformance with 3.3.3.2, following any discharge rate. The requirement may be demonstrated in conjunction with 4.6.3.2.

4.6.3.4 Hydrogen evolution. Each cell shall be tested for hydrogen evolution on open circuit to determine conformance with 3.3.5. The open circuit voltage period shall start after the end of a charge conducted in accordance with the manufacturer's service manual. Gas collection shall commence at the beginning of the open circuit period and quantities shall be periodically recorded until completion of the test.

4.6.3.5 Angle of operation. Each cell, within 1 hour after ending any charge, shall be tilted in any direction 45 degrees from the vertical to determine conformance with 3.5.3. The cells shall be tilted, for information only, to 60 degrees from the vertical or to an angle less than 60 degrees where spill occurs, to determine the maximum inclination capability of the manufacturer's design. The data shall be recorded.

4.6.3.6 Electrolyte level indicators. Functional use of electrolyte level indicators, primary and alternate, shall be demonstrated on actual cells for various levels which occur during charged, discharged, and open cirucit states or which may occur during abnormal conditions (too high or too low) to demonstrate the level requirement of 3.4.9.3.

4.6.3.7 Terminal temperature. Cells shall be connected as a battery and insulated on 4 sides so as to approximate an adiabatic condition as closely as possible. The temperatures of the positive terminal and electrolyte shall be measured by thermocouples or thermistors and recorded at 15-minute intervals during a 275-A rate discharge with an electrolyte starting temperature and an ambient temperature in the range of 16°C to 32°C. The cells shall meet the requirement specified in 3.3.4.

4.6.3.8 Flash arrester tests.

4.6.3.8.1 Dry test. Air shall be vented through each dry flash arrester at the rate of 1 liter per minute. The pressure drop of the air passing through the flash arrester shall be measured with a water manometer and shall meet the requirements specified in 3.4.12.2.1.

4.6.3.8.2 Wet test. Each flash arrester shall be submerged in 45 percent ± 1 percent KOH for 2 minutes at an alkali temperature of 25°C ± 3°C. No sooner than 15 seconds nor later than 60 seconds after removal from the alkali bath, air shall be vented through the flash arrester at the rate of 1 liter per minute. During this period, the flash arrester shall be in the normally installed position to permit drainage. The pressure drop of the air passing through the flash arrester shall be measured with a water manometer, and shall meet the requirements specified in 3.4.12.2.2.

4.6.3.8.3 Flash arrestment test. While a gas mixture consisting of 67 percent hydrogen and 33 percent oxygen is passed through each dry flash arrester from the inside at the rate of 1 liter per minute, an electric spark shall be generated continuously within 15 mm of the outside of the flash arrester for 15 seconds. Flash arrestment ability shall meet the requirements specified in 3.4.12.3.

4.6.3.9 Failure procedure. In case of failure, the tests of 4.6.3 may be repeated once. In case of subsequent failure, the first article cells which fail to meet individually the tests of 4.6.3 shall be rejected and may not be offered again for delivery unless a proposal for the correction of the deficiency and retest of first article cells has been submitted to and reviewed by NAVSEC.

4.6.4 Inspection of wet sample cells and wet battery.

4.6.4.1 Cell formation. Sample and battery cells shall receive a minimum of 2 formation discharges before any test cycling. Each cell, during each formation change, shall be

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individually charged as necessary to obtain a cell voltage of at least 2.0 volts. The following shall be a part of the formation cycling:

(a) Vacuuming of each cell shall be performed at least:
  (1) After initial filling with electrolyte.
  (2) At the end of the soaking period.
  (3) At the end of each formation charge.
(b) Adjust electrolyte, if required, after each vacuuming.
(c) For each individual cell, the quantities of electrolyte that are used for initial filling and are added or removed from a cell during formation shall be measured and recorded.

4.6.4.2 Resistance test. The internal resistance of each sample and battery cell, in a discharged state after formation, shall be measured between terminals of opposite polarity. The resistance shall be measured with a J.W. Dice Co., model 150 or 153 microohmmeter, Keithley Instrument Inc., model 503-30 milliohmmeter, or equal. Cells shall meet the requirements of 3.3.7.

4.6.4.3 Inspection of wet sample cells. After formation (see 4.6.4.1), each group of sample cells, selected in accordance with 4.4.1, shall be series-connected as a 6-cell battery and shall be submitted to the cognizant DCAS to witness the acceptance performance cycle. This cycle shall consist of a full charge in accordance with the manufacturer's service manual instructions and a full discharge at 170A. Each cell shall, during the discharge, be individually removed from the discharge circuit when the cell reaches 1.31 volts. Data on voltage, current, time, and electrolyte level at the end of charge and discharge shall be monitored and recorded for each cell. The results of the capacity discharge test for each cell shall be averaged and the estimated standard deviation calculated according to the following formula:

$$s = \left(\frac{\sum_{1}^{n} (x_i - \bar{x})^2}{n-1} \right)^{1/2}$$

Where:
  s = Estimated standard deviation.
  n = Number of sample cells.
  $\bar{x}$ = Average of results of tests.
  X_i = Result of individual sample cell.

The capacity and voltage of each cell shall meet the requirements of 3.3.1 and the average percent of rated capacity for all cells shall be equal to or greater than 100 percent of rated capacity plus 2.45 s. No more than 2 cycles are permitted to demonstrate conformance with this requirement. Adjustment of electrolyte level or individual cell charging during or after the acceptance cycle shall invalidate that cycle.

4.6.4.4 Notification to ship battery. Upon written notification by NAVSEA, the contractor shall be required to ship the battery dry, or to fill and ship the battery wet, or to fill and form the battery prior to shipment. In the latter case, the contractor shall demonstrate performance as specified in 4.6.4.1, 4.6.4.2 and 4.6.4.5.

4.6.4.5 Inspection of wet battery. After formation (see 4.6.4.1), the 150 cells shall be series-connected as a battery and submitted to the cognizant DCAS to witness the acceptance performance cycle. This cycle shall consist of a full charge in accordance with the manufacturer's service manual instructions and a full discharge at 170 A to the cut-off voltage. No more than 2 cycles are permitted to demonstrate 100 percent rated capacity in accordance with the voltage requirements of table I. Electrolyte levels of all cells shall be recorded after discharge and shall be within the levels set by the manufacturer's service manual. Adjustment of electrolyte level or individual cell charging or discharging during or after the acceptance cycle shall invalidate that cycle.

4.6.4.6 State of battery cells for delivery. After demonstration of performance, the cells shall be recharged prior to delivery and shipment, unless otherwise directed by NAVSEA. The Ah input and individual cell electrolyte levels shall be recorded after recharge. The input shall be at least 90 percent of the previous output and the electrolyte levels of all cells shall be within the levels set by the manufacturer's service manual.

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4.7 Shipment of test cells. After completion of the tests of 4.6.3 on the first article cells (see 4.3) and the sample cells (see 4.4.1), all these cells shall be shipped in a discharged state to a Government laboratory as specified (see 6.2.1) to be subjected to life testing as specified in 4.8.

4.8 Life testing. The cells received in a Government laboratory (see 4.7) shall be tested for life over 24 months on several routines as specified and when instructed by NAVSEC to verify conformance with section 3 of this specification.

4.9 Disposition of used cells. The used cells, after retest (see 4.6.3.9), may be submitted to analysis at NAVSEC's discretion prior to shipment to the Government silver reclamation activity as directed by NAVSEA (see 6.2.1).

4.10 Test procedures. The contractor shall prepare a test procedure to cover first article and quality conformance inspections (see 4.2) and a test procedure to cover inspection of the wet battery (see 4.6.4) in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.11 Test reports.

4.11.1 First article inspection report. The contractor shall prepare a first article inspection report (see 4.3) in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.11.2 Sample cell (wet condition) test report. The contractor shall prepare a sample cell test report in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.11.3 Wet battery test report. After formation of the battery cells (see 4.6.4.1), the contractor shall prepare a summary of the battery performance in accordance with the data ordering document included in the contract or order (see 6.2.2), and shall include the following:

(a) Cell formation data (see 4.6.4.1).
(b) Capacity inputs and outputs.
(c) Cut-off voltages for the battery and end-voltage of each cell.
(d) Average discharge voltage for the battery.
(e) Individual cell electrolyte levels after last discharge and last charge.

4.11.4 Silver metal accountability report. A report on accounting of silver metal furnished by the Government and used in the cells delivered shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2). The report shall include the amount of silver received by the contractor, the amount per cell, for active material and for inactive material (grid and plating), and the total amount used in the cells delivered.

4.12 Inspection of preparation for delivery. Preservation-packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. *PREPARATION FOR DELIVERY*

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.6.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Cells (see 3.6.1). Cells shall be individually unit protected method III in accordance with MIL-P-116 and packed as specified in 5.2.

5.1.1.2 Cell markers (see 3.4.14.1), repair parts (see 3.6.2) and accessories (see 3.6.3). Cell markers, repair parts, and accessories shall be segregated and each item in the quantities required for use, except as specified herein, shall be unit protected

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method III in accordance with MIL-P-116 in individual paperboard folding or set-up cartons or fiberboard boxes conforming to PPP-B-636, domestic grade.

(a) Cell jars - One to a fiberboard box.
(b) Cell covers - Each cover shall be wrapped or interleaved with a resilient cushioning material.
(c) Flash arresters - One to a paperboard carton. Each arrester shall be wrapped or cushioned with a resilient material. For repair parts, ten cartons shall be intermediate packaged in each of 3 fiberboard boxes.
(d) Level indicators - Viewing surface shall be covered with a non-abrasive wrap and secured with pressure-sensitive tape. The lubricated O-ring and adjacent areas shall be covered with a waterproof, greaseproof wrap and secured with pressure-sensitive tape. Indicator stem shall be inserted in a plastic or paperboard tube to provide rigidity and prevent breakage. Each indicator shall then be wrapped with a resilient cushioning material and each indicator placed in a paperboard folding or set-up carton or in partitions and then for repair parts consolidated in a fiberboard container.
(e) Electrolyte - Electrolyte shall be packaged in 3 to 4 liter plastic bottles conforming to the level A requirements of PPP-C-300.

5.1.1.2.1 Accessory sets. When specified (see 6.2.1) accessories, (see 3.6.3) shall be packaged in individual sets in the quantity and type required as follows:

(a) One each cell cover, flash arrester and level indicator packaged as specified in 5.1.1.2 shall be placed inside a cell jar. Cushioning or blocking shall be applied to prevent movement of the packaged parts within the jar.
(b) Loose fill cushioning (see 5.3) shall not be used.
(c) Each cell jar shall be packaged as specified in 5.1.1.2 (a).

5.1.2 Level C.

5.1.2.1 Cells. Cells shall be packaged as specified in 5.1.1.1.

5.1.2.2 Cell markers, repair parts and accessories. Cell markers, repair parts and accessories shall be segregated and each item, in the quantity required for use, shall be packaged in a manner which will afford protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity for immediate use. The contractor's normal preservation-packaging methods may be used when such meets the requirements of this level.

(a) Electrolyte - Electrolyte shall be packaged in 3 to 4 liter plastic bottles conforming to the level A requirements of PPP-C-300.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2.1).

5.2.1 Level A. Cells, cell markers, repair parts, accessories, and electrolyte shall be packaged as specified in 5.2.2 except the containers shall be of the overseas type or class.

5.2.2 Level B.

5.2.2.1 Cells. Cells shall be packed 10, except spare cells shall be packed 9, to a shipping container. Containers shall be designed using PPP-B-601 and PPP-B-621 for boxes, and PPP-C-650, MIL-C-3774 and Department of Transportation Code of Federal Regulations, Title 49, section 173,260 subpart E for crates, and as follows:

(a) Containers shall accept either filled or dry cells. Each filled cell shall be placed in a plastic bag to entrap any electrolyte spilled during shipment.

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(b) Cells shall be segregated within the container by separators or partitions.
(c) Container sides and ends shall be modified by the addition of structural members to prevent container distortion.
(d) Container tops shall be secured by screw or screw-type fasteners in lieu of nailing.
(e) Containers shall have a minimum of two 80 by 100 mm (nominal) wood skids.
(f) Containers shall be designed for reusability.

5.2.2.2 Cell markers, repair parts, and accessories (less electrolyte). Material, packaged as specified, shall be packed in fiberboard boxes conforming to PPP-B-636, domestic grade or PPP-B-640, class 1. Alternatively, containers shall conform to PPP-B-601, domestic type; PPP-B-621, class 1; PPP-B-576, class 1; or PPP-B-585, class II. Container and style selections shall be at the option of the contractor. Container closure and reinforcing, as applicable, shall be in accordance with the applicable container specification or appendix thereto.

5.2.2.2.1 Accessory sets. Accessory sets packaged as specified in 5.1.1.2.1 shall be packed four sets to a shipping container. Shipping containers shall conform to 5.2.2.2 or 5.2.1 as applicable for the level of packing specified (see 6.2.1).

5.2.2.3 Electrolyte. Electrolyte, packaged as specified, shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, as specified in Department of Transportation Regulations, Title 49, section 173.260, subpart E, gross mass shall not exceed 30 kg. Alternatively, electrolyte packaged as specified shall be packed in wood containers specified in 5.2.2.2 with padding and separators between the electrolyte bottles and inside faces of the wood containers.

5.2.2.4 Consolidated packing. Items specified in 5.2.2.2, when supplied with a complete battery, may be consolidated within a single shipping container. The container(s) shall be provided with 2 index lists completely itemizing and identifying the container contents. One copy shall be placed inside the container in a location offering ease of access and removal. One copy shall be attached to the exterior side or end of the container in a protected location which will prevent loss and damage during shipment and storage. Lists shall be enclosed in a waterproof sealed bag or envelope.

5.2.3 Level C.

5.2.3.1 Cells. Cells shall be packed as specified in 5.2.2.1.

5.2.3.2 Cell markers, repair parts, accessories, and electrolyte. Cell markers, repair parts, accessories, and electrolyte, packaged as specified, shall be packed in containers acceptable to the common carrier which shall insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight or National Motor Freight rules and regulations or other carrier regulations as applicable to the mode of transportation.

5.3 Cushioning dunnage and wrapping materials.

5.3.1 Level A preservation-packaging and levels A and B packing. Use of all types of loose-fill materials for packaging and packing applications such as cushioning, filler, dunnage or blocking (see 5.1.1.2.1), is prohibited for material destined for shipboard installation/stowage.

5.3.2 Level C preservation-packaging and packing. When loose-fill type materials are used for packaging and packing applications such as cushioning, filler and dunnage, all containers (unit, intermediate and shipping) shall be marked or labelled with the following information:

"CAUTION

Contents cushioned etc., with loose-fill material.
Not to be taken aboard ship.
Remove and discard loose-fill materials before shipboard stowage.
If required, recushion with cellulosic material, bound fiber, fiberboard or transparent flexible cellular material."

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5.3.3 Cushioning, filler, dunnage and wrapping materials selected, whenever available, shall exhibit improved preformance for resistance to fire.

5.4 Marking. In addition to any special marking required (see 6.2.1), or herein, interior packages (unit and intermediate) and shipping containers shall be marked in accordance with MIL-P-116 and MIL-STD-129 as applicable to the levels of packaging and packing specified. In addition, unit packages and shipping containers of electrolyte shall be marked or labelled in accordance with the applicable Department of Transportation requirements as specified for hazardous material.

5.4.1 Special markings.

5.4.1.1 Cell containers. Cell containers shall be marked with the following:

(a) "DO NOT DESTROY - REUSABLE."
(b) "FRAGILE" and up-arrows (see MIL-STD-129).
(c) "STORE IN COOL AREA" (preferably minus 7°C to 4°C) and other storage criteria deemed necessary by the contractor for guidance to the receiving and storing activity for filled and spare/empty cells.
(d) Labelling as required by Department of Transportation.
(e) Structural and handling markings such as center-of-balance and lift points.
(f) Mercury (see 3.4.15).

5.4.1.2 Accessory sets. Unit and shipping containers (see 5.1.1.2.1 and 5.2.2.2.1) in addition to any special marking required, shall be marked as follows:

(a) Unit containers
    1. NR-1 cell parts.
    2. Set A or Set B as determined by the type level indicator included.
    3. NICN number (see 6140-LL-HAC-____)
(b) Shipping containers.
    1. NR-1 cell parts.
    2. Quantity of Set A or Set B as applicable.
    3. NICN number (example 6140-LL-HAC-____).
    4. Box _____ of _____ boxes (to identify total of containers in shipment, when applicable).

(Note. refer to MIL-STD-129 for application of set markings.)

5.5 Technical data (see 3.9, 4.10, and 4.11). When specified (see 6.2.1 and 6.2.2), packaging-packing drawings shall be submitted. These drawings are not required when previously submitted and accepted. Drawings, service manuals, test procedures and reports furnished to the procuring activity shall be prepared for shipment in accordance with level C requirements (see 5.1.2 and 5.2.3).

6. NOTES

6.1 Intended use. This specification covers silver-zinc battery cells for use on NR-1 submarines.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Use and amount of mercury or mercury substitute and identification of mercury substitute (see 3.4.15.1).
(c) Address for shipment of the battery cells (see 3.6.1).
(d) Total number of cells to be delivered.
(e) Address for shipment of the spare cells (see 3.6.1).
(f) Address for shipment of repair parts (see 3.6.2).
(g) Address for shipment of accessories, if accessories are required (see 3.6.3).
(h) If engineering services are required (see 3.7).
(i) Address for shipment of first article and sample cells to a Government laboratory (see 4.7).

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(j) Disposition of used cells (see 4.9).
(k) Level of preservation-packaging and methods required (see 5.1).
(l) When accessories are to be furnished in sets (see 5.1.1.2.1).
(m) Level of packing and methods required (see 5.2).
(n) Special marking required (see 5.4).
(o) When packaging-packing drawings are required (see 5.5).

6.2.2 Data requirements. When this specification is used in a procurement which invokes the provision of the "Requirements for Data" of the Armed Services Procurement Regulations (ASPR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the ASPR are not invoked in a procurement, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID	Options
3.7	Report, contractor engineering and technical services (CETS)	UDI-E-23127	-------
3.9.1	Drawings, engineering and associated tests, level 3 (production)	DI-E-7015	Drawing No. - Contractor Design activity designation - Contractor
3.9.2	Technical publications for advanced development programs	DI-M-6157	-------
3.9.3	Plan, milestone (type II) for critical supply contracts	UDI-A-23015	-------
3.9.3	Report, milestone (type II)	UDI-A-23018	-------
4.1.1	Quality program plan	UDI-R-23743	-------
4.10	Test procedures	UDI-T-23649	-------
4.11.1	Report, first article inspection	UDI-T-23450	-------
4.11.2, 4.11.3, 4.11.4	Test reports	DI-T-2072	------- .
5.4	Drawing, preservation-packaging, packing, and marking (PPP&M)	UDI-E-23123	Category A

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 or the specification required to be performed to meet a data requirement may be waived by the procuring/purchasing activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item procured to this specification. This does not apply to specific data which may be required for each procurement regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Guarantee. The guarantee shall be as specified in 6.4.1 through 6.4.4.

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6.4.1 Notwithstanding inspection and acceptance by the Government of supplies furnished under the contract or any provisions of this contract concerning the conclusiveness thereof, the contractor guarantees the performance of the battery in terms of service operating conditions (months of service or total Ah output). The battery shall be guaranteed for 24 months or to deliver a total output of 50,000 Ah, whichever occurs first, before the capacity drops to the percentage of rated capacity as specified herein, when tested at any rate in table I and subject to the temperature limitations of this specification. The period of the guarantee shall begin on the date the battery is filled with electrolyte but not later than 24 months after delivery.

6.4.2 The battery shall be considered to have failed when the battery or at least 9 cells (maintained and handled in accordance with the usual methods specified by the supplier in the service manual) reach the conditions described below. The use or availability of spare cells for replacement of failed cells in the battery shall not alter the terms of this guarantee. Any cell shall be considered failed in the following cases:

(a) The cell is internally shorted.
(b) The cell limits the battery capacity to a percentage below those specified in 6.3.3 and 6.3.4 and has to be removed in order to raise overall battery capacity.
(c) The cell develops leaks through cracks, thus causing grounds in the ship.

6.4.3 Until the end of the first 12 months following electrolyte filling or until the battery delivers an output of 25,000 Ah, whichever occurs first, the capacity shall be at least 90 percent of the rated capacity as shown in table I. If the battery fails to meet these requirements, during these conditions of life or accumulated Ah output, the contractor will replace the total battery at no cost to the Government.

6.4.4 For any condition after those stated in 6.4.2 and 6.4.3 in terms of life or accumulated Ah output, and prior to the end of the guaranteed life or prior to delivering the total required output of 50,000 Ah, whichever occurs first, the capacity shall be at least 80 percent of the rated capacity shown in table I. In case of failure to meet these requirements during these conditions, the contractor shall indemnify the Government on a prorata basis (proportionately) for months of service not obtained or total Ah output not obtained, whichever is smaller. (If P is the purchase price of the battery, n the number of months not obtained, Q the total Ah output not obtained, the smaller of the 2 numbers $P\frac{n}{24}$ or $P\frac{Q}{50,000}$ shall be used.)

6.4.5 The same guarantee requirements of life, total Ah output, minimum capacity and indemnification specified in 6.4.1, 6.4.3, and 6.4.44 shall apply to each spare cell individually, whether wet or dry.

6.5 Total number of cells to be delivered. The cells to be delivered are divided as follows:

(a) 6 cells for first article inspection for the entire contract (see 4.3).
(b) 6 cells as samples for quality conformance inspection for each 150-cell battery (see 4.4.1).
(c) 150 cells for each battery (see 3.6.1).
(d) 9 cells as spares for each 150-cell battery (see 3.6.1).

6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Changes from previous issue. The margins of this specification are marked with "#" to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - SH
(Project 6140-N468)

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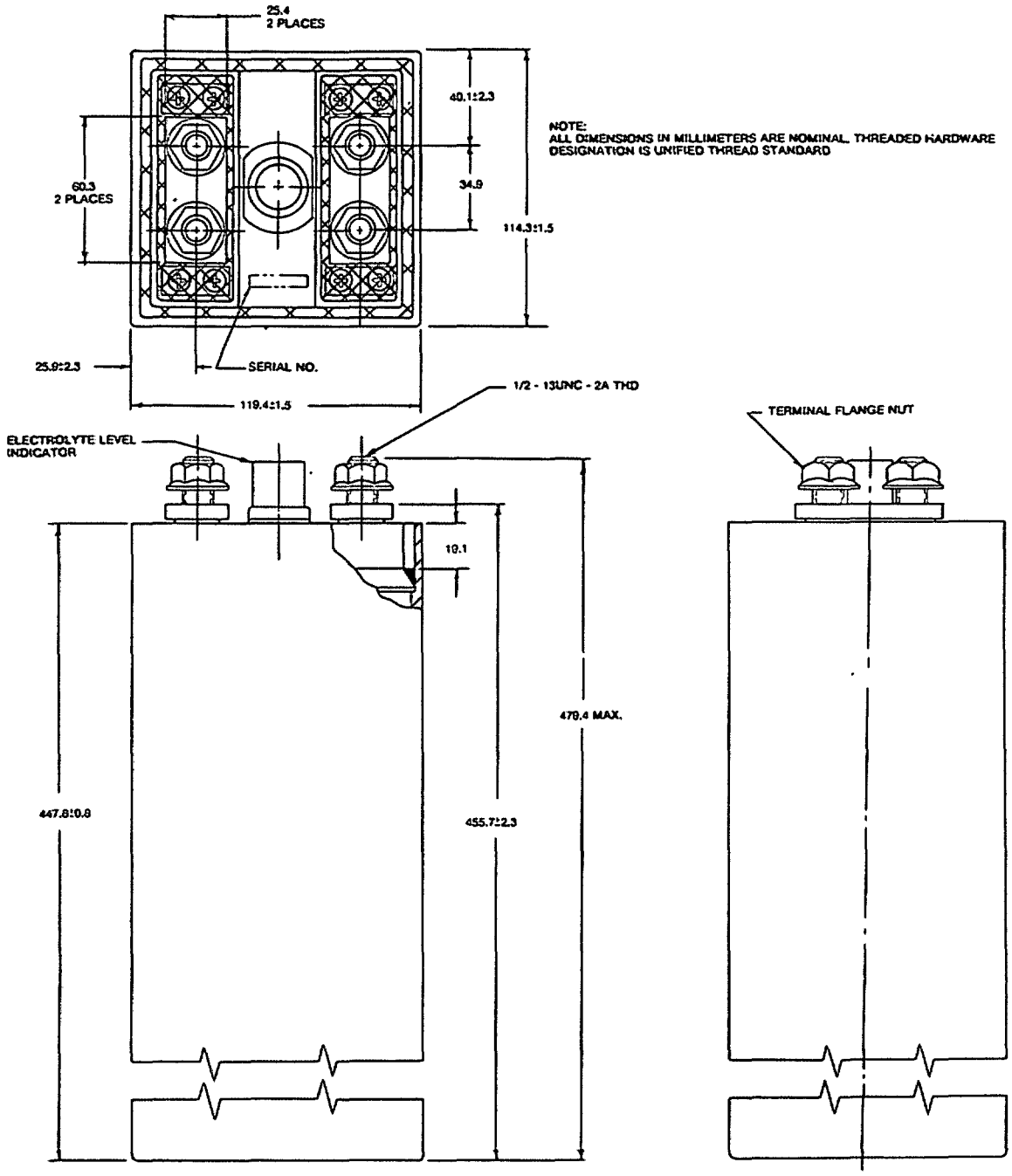
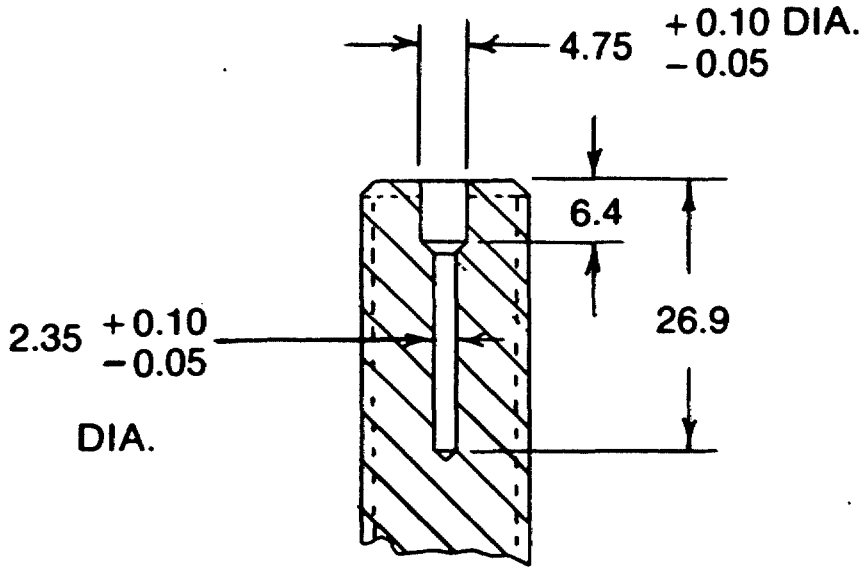


FIGURE 1. Cell outline

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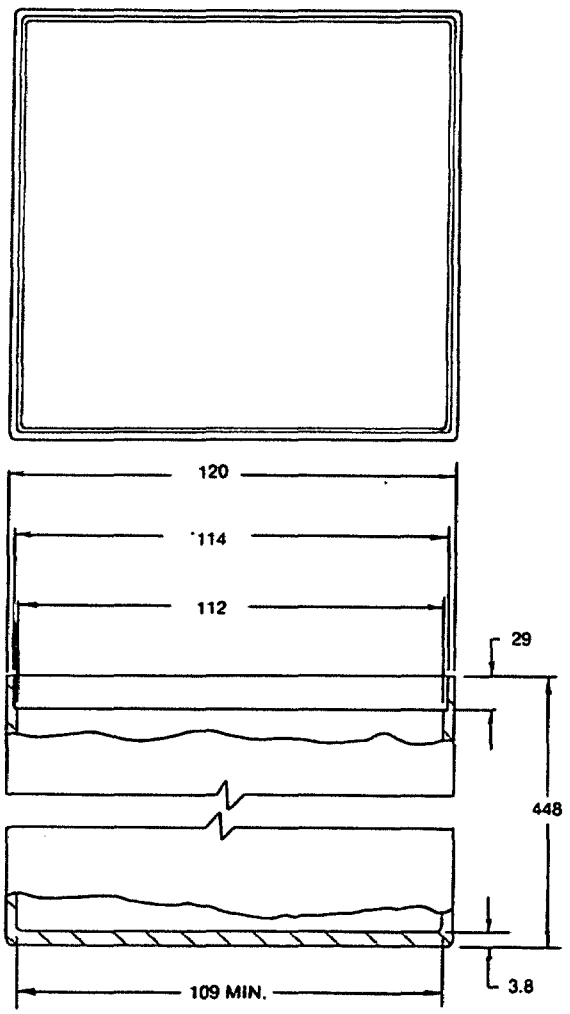


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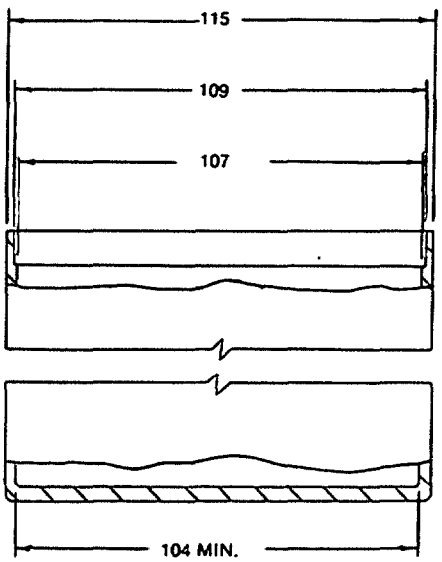
NOTE:
DIMENSIONS ARE IN MILLIMETERS.

FIGURE 2. Temperature monitoring hole.

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NOTES:
1. ALL DIMENSIONS IN MILLIMETERS ARE NOMINAL.
2. STRESS RELIEVE IN AIR AT 165°C±3°C FOR AT LEAST 4 HOURS.



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FIGURE 3. Cell jar.

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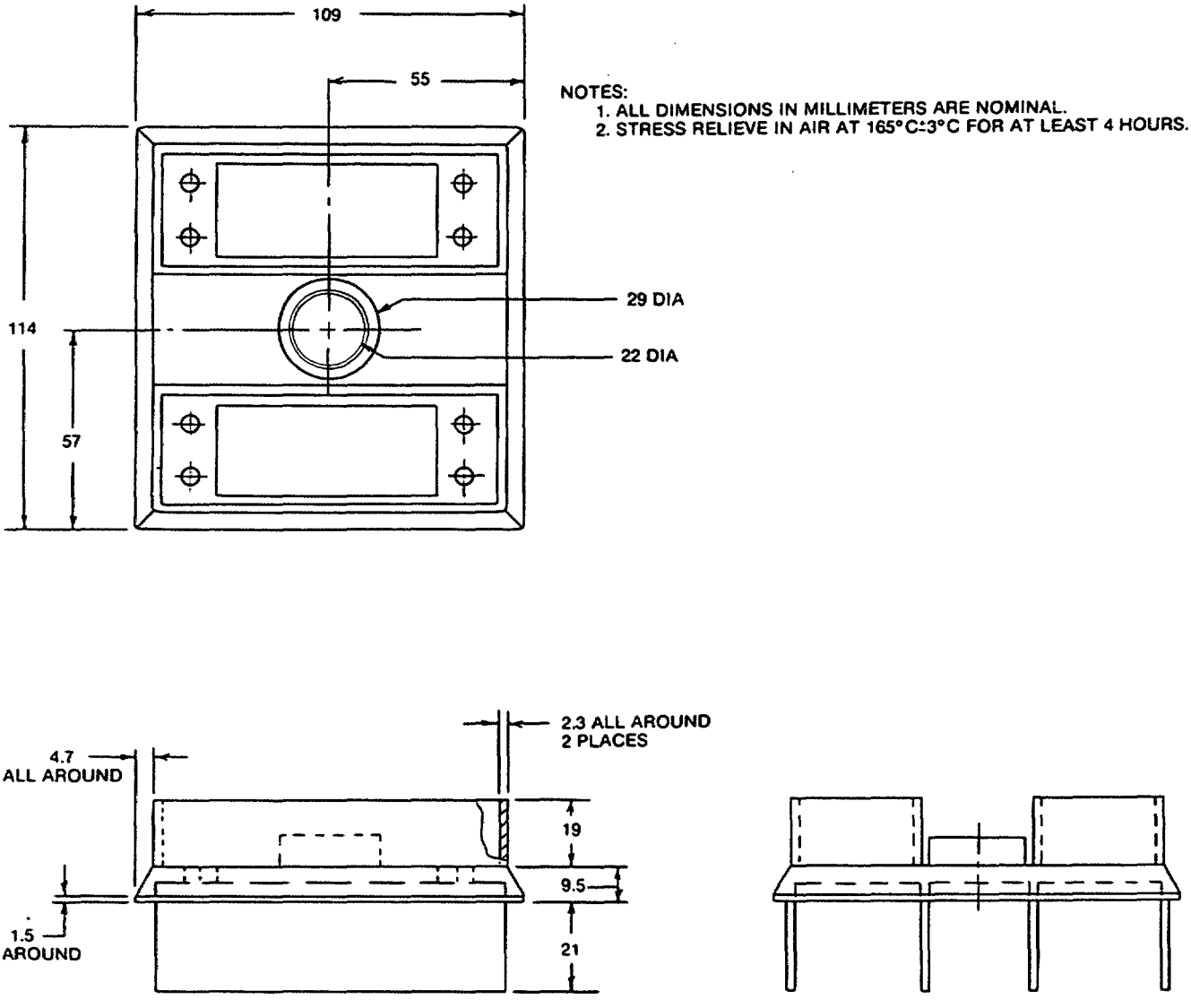


FIGURE 4. Cell cover.

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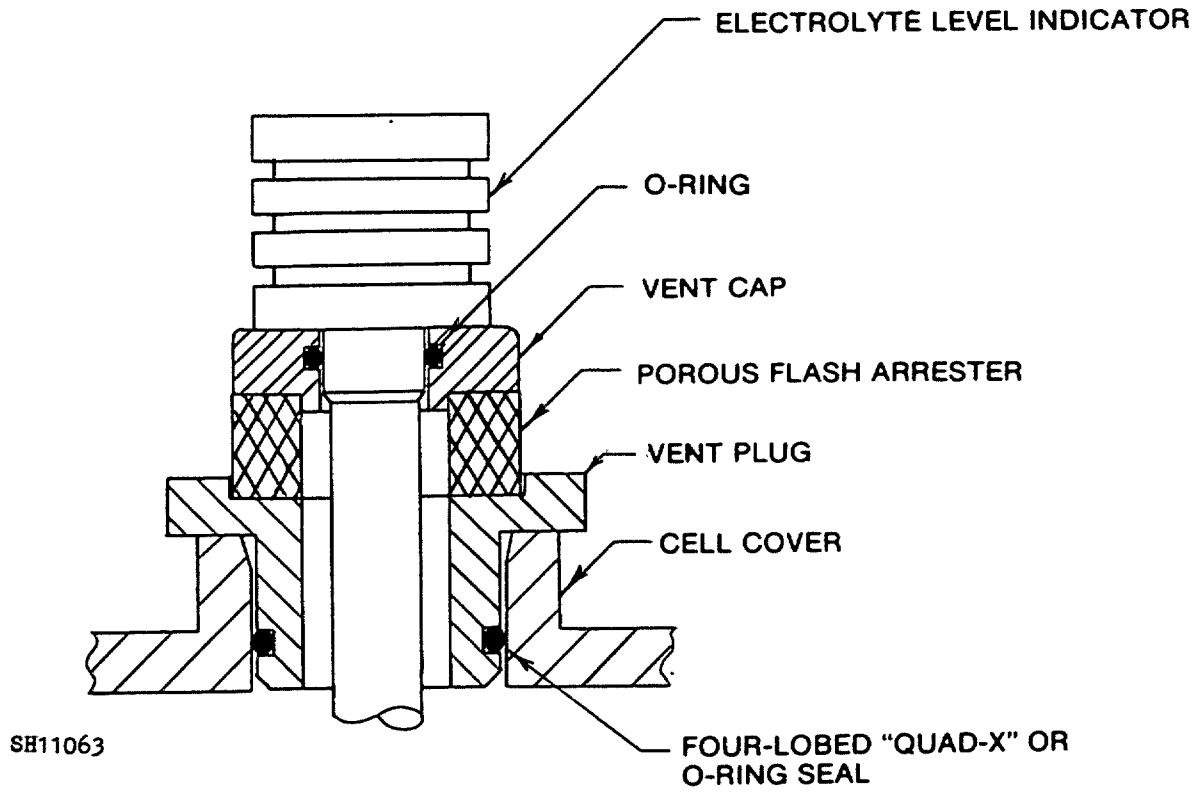


FIGURE 5. Electrolyte level indicator and flash arrester assembly.

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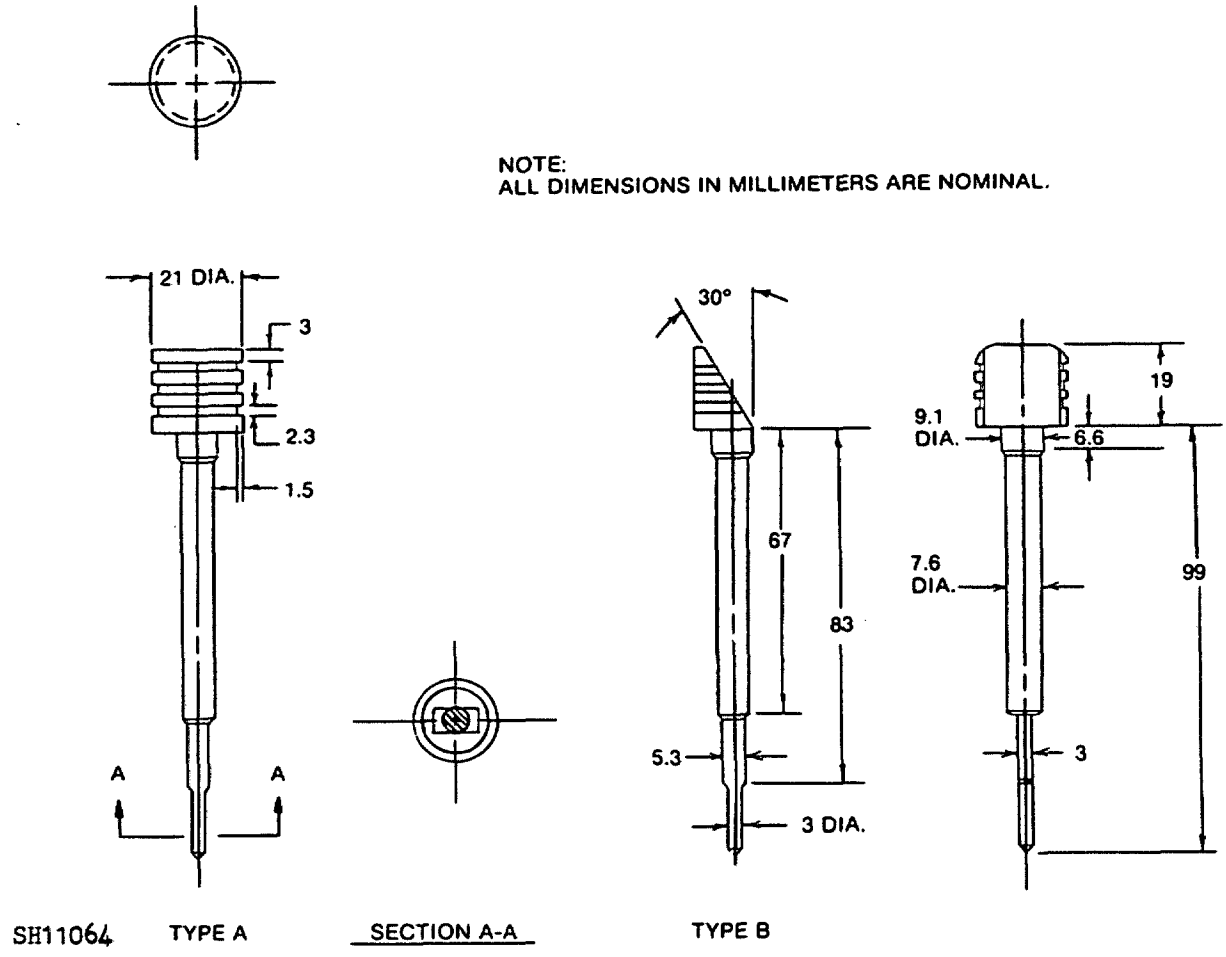


FIGURE 6. Electrolyte level indicator.

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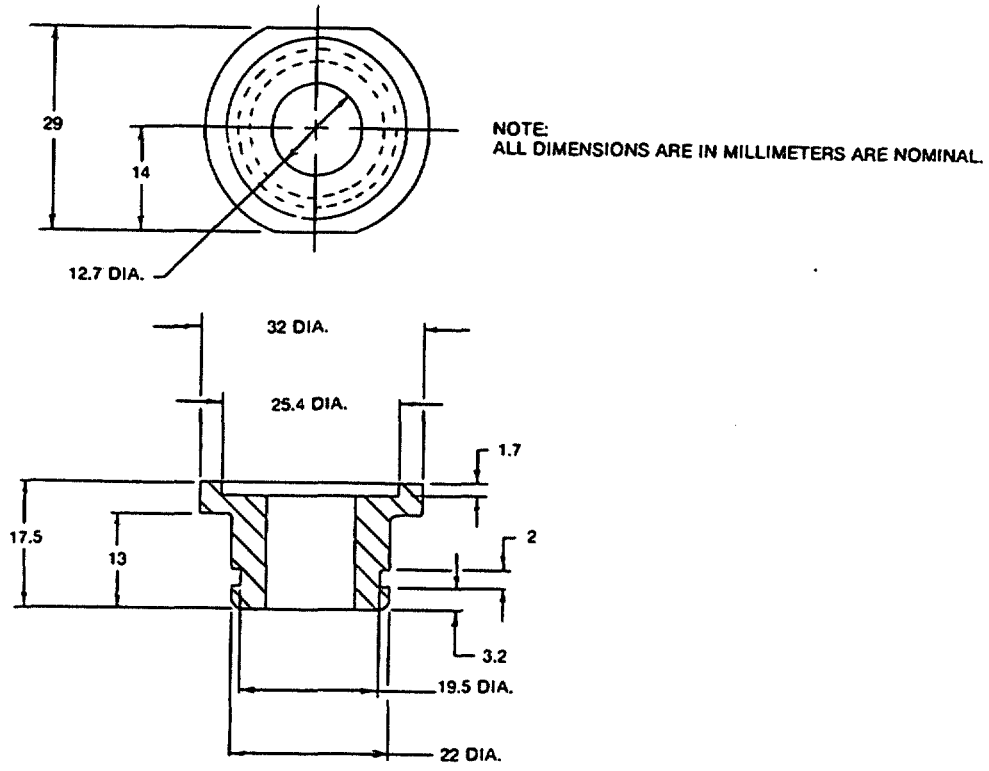


FIGURE 7. Flash arrester assembly.

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APPENDIX

PREPARATION INSTRUCTIONS FOR SILVER-ZINC BATTERY SERVICE MANUAL

10. SCOPE AND OBJECTIVE

10.1 Scope. This appendix covers the instructions which apply to the preparation of a service manual for the silver-zinc type of battery used in Navy submersibles. The manual is provided by the battery manufacturer to describe the required procedures for installation and operation of the specific battery design identified in the manual.

10.2 Objective. The instructions for the preparation of a battery service manual have several objectives as follows:

(a) Prevent the omission of necessary information by identifying the general subjects to be covered.
(b) Improve the usefulness of the manual by establishing a basic plan for the presentation of the manufacturer's instructions in a fixed sequence under specified general subjects.
(c) Prevent the use of an obsolete issue of a service manual by establishing a method of identifying the issue and date of each manual in a manner easily recognized by the users. Requirement for the preparation of a new service manual (with a new identification number supplied by the Navy) for each new battery contract is also intended to aid identification of the appropriate service manual for each battery design.

20. APPLICABLE DOCUMENTS

20.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this appendix to the extent specified herein.

SPECIFICATION

MILITARY
MIL-M-15071 - Manuals, Technical; Equipment and Systems, Content Requirements for.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 General. The instructions shall be considered as a supplement to the requirements of MIL-M-15071 and shall take precedence only when a specific requirement is identified herein.

30.2 Cover. The cover shall be hard cover containing the information shown on example A in accordance with the following and in bold, large characters:

(a) Navy publication number as provided after acceptance of the manual.
(b) Navy's designation for the battery.
(c) Navy identification of the ship.

30.3 Title page (first page). The title page shall contain the information as shown in example B in accordance with the following:

(a) Navy publication number as provided after acceptance of the manual.
(b) Navy's designation for the battery.
(c) Navy identification of the ship.
(d) Battery manufacturer's name as stated in the contract.
(e) Change number and dates of original and of changes to the manual.

30.4 List of effective pages. The list of effective pages shall be prepared as shown on example C.

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30.5 Change record. A permanent change record shall be prepared as shown on example D.

30.6 Approval and procurement record. An approval and procurement record page shall be prepared as shown on example E.

30.7 Content assurance. A content assurance record shall be prepared as shown on example F.

30.8 Table of contents. The table of contents shall be organized as shown on example G. The pages shall be consecutively numbered and the page numbers reflected in the table of contents. The titles shall appear as shown and also in the order listed.

30.8.1 Chapters. The manual shall be arranged on the basis of eight chapters.

30.8.2 Section. Certain chapters, as indicated, shall be divided into sections as follows to provide an orderly grouping of related articles:

(a) Sections within each chapter shall be lettered sequentially as A, B, C, etc. prior to the title.
(b) Each section number with title and the corresponding page number shall appear in the table of contents.

30.8.3 Articles. Each chapter shall be subdivided into numbered articles as follows:

(a) Each article shall be identified by a three digit number, with the first digit identical to the chapter number.
(b) Each article shall have a title which is also listed on example G.
(c) Where appropriate, articles may be further subdivided using "a., b., c., and d., etc." for identification prior to the title.

30.8.4 Uniformity. Each manual shall be organized and written to maintain uniformity in the presentation of the information and instructions for the installation and operation of silver-zinc batteries. Wording of the same article shall be identical in all manuals when the intent of the information or the action to be taken is identical for different batteries. If a standard subject for an article as identified on example G is not appropriate for a specific battery design, the text of the service manual for that article shall state "Not Applicable". If a required subject is not included in the standard table of contents, an article with number and title shall be added to the appropriate chapter or section.

30.8.5 Additional text requirements. The following requirements apply to the text of the article indicated:

(a) 212-Cell cover. Include "cover support" and "cover-to-case sealant" discussions.
(b) 217-Flash arresters. Include relationship to electrolyte spillage and cross reference discussion of spill angle limitation in chapter 4, if applicable.
(c) 223-Electrolyte. Identify the KOH concentration and the percentage of impurities present.
(d) 224-Oil. Identify function in pressure compensated cells and the characteristics of oil used.
(e) 245-Hydrogen detector. Identify portable and installed types as appropriate.
(f) 248-Discharge equipment. Identify battery test discharge and shorted cell devices as appropriate.
(g) 262-Voltage. Include specific discussions of voltage on charge, discharge, float, open circuit and the related average, plateau, and limits as appropriate.
(h) 311, 312, and 313-Storage. Identify specific requirements for cells and battery as appropriate.
(i) 321-Safety precautions. This article should cross reference to chapter 8.
(j) 324-Formation. Identify specific requirements applicable to articles 311, 312, and 313 as appropriate.
(k) 326-Oil filling. Include instructions for dye to coloring of oil.
(l) 336-Installation. Procedure should apply to individual cells or assembled battery as appropriate.

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(m) 401-Safety precautions. This article should cross reference to chapter 8.
(n) 402-Auxiliary equipment. This article should cross reference to chapter 2, section C.
(o) 403-Jumpered cells. Include instructions for maintenance of good cell disconnected from one battery to balance with jumpered cell in parallel battery as appropriate.
(p) 418-Maintenance charge. Provide instructions as to when to be used.
(q) 420-Voltage limits. Specify the required voltage accuracy.
(r) 424-Charging procedures. Include instructions for parallel batteries as appropriate.
(s) 436-Capacity test discharge. Include instructions for parallel batteries as appropriate.
(t) 504-Maintenance schedule. Include a chart or table with discussion of requirements.
(u) 511-Voltage. Specify the accuracy with which the various cell voltage (charged stand open circuit, charge, discharge and float) can be identified.
(v) 515-Meters. Cross reference to chapter 2, section C.
(w) 519-General cleanliness. Topics should include flash arresters, bubble breakers, cell covers, encapsulating material, battery box (drain and flushing instructions specified) and battery compartment as appropriate.
(x) 521-529-Shipboard action. Instructions in these articles should support the standards identified in articles 511 through 519. Shipboard capabilities may be limited due to battery access for a specific ship system. Required frequency of measurements, particularly for cell voltage shall be clearly stated.
(y) 531-539-Support ship action. Instructions in these articles should also support the standards identified in articles 511 through 519. Support ship requirements must be identified in conjunction with shipboard capabilities which are directly related to battery access for specific ship system.
(z) 541-General. Include in this article that these recommendations are intended to prolong battery life. Also any action taken based upon these recommendations is subject to not conflicting with ship operations.
(aa) 543-Booster charge. Instructions for use following prolonged open circuit stand as appropriate.
(bb) 544-Temporary storage. Instructions for battery maintenance during extended periods when it is planned not to require battery operation.
(cc) 545-Low temperature. Instructions concerning the advantages of battery storage at low temperatures and the appropriate procedures.
(dd) 546-Electrolyte level. If appropriate, identify the optimum procedure for maintaining the required electrolyte level.
(ee) 547-Characteristics. Characteristics of each battery design should be reviewed to identify additional recommendations.
(ff) 602-Internal shorting. Identify the characteristics of cells with a "slow short" versus a "hot short".
(gg) 612-614-Float, equalizing and maintenance charges. Include discussion of why capacity loss is minimized by each of those charging procedures. Provide cross references to prior discussions in articles 416, 417 and 418 respectively as appropriate.
(hh) 621-622-Detection and verification. Specific instructions in each of these articles related to (1) open circuit charged stand, (2) during recharge, (3) on float, and (4) during discharge as appropriate shall be provided.
(ii) 623-Action. Provide separate instructions for "slow short" and "hot short".
(jj) 624-Troubleshooting diagram. Provide a diagram with appropriate explanation as described in MIL-M-15071. Base diagram on the detection, verification and actions required as appropriate.
(kk) 631-When to replace. Discuss failed cell (shorted) and limiting cell (low capacity).
(ll) 632-How to replace. Cross reference to chapter 7, section 3.
(mm) 701-704-Seals. Discuss the various types of sealing (pre-mixed and prepared by user materials and mechanical seals such as gaskets or "O" rings) as appropriate in each article. Precautions in handling the materials shall be clearly identified.
(nn) 732-734-Replacing cells. Include appropriate article cross references to chapter 3, sections C and D.

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(oo) 741-General. Identify possible causes for low capacity and cite cross references to articles in chapter 2, section D, as appropriate.
(pp) 742-Procedure. Describe the procedure related to each cause discussed in article 741. Cross reference the appropriate article under charging (chapter 4, section B) or cell jumpering (article 403) and cell replacement (chapter 7, section D) as appropriate.
(qq) 801-General. Identify whether any of the recommended safety equipment is supplied by the battery manufacturer.
(rr) 802-809-Equipment. Describe the item, its common name, manufacturer's name and part number if a particular type is either recommended or the only one available and discuss the purpose. Also provide procedure for use of special devices and cite cross references to other appropriate articles for all items. Article 807 should identify materials such as an eye wash bottle, vinegar, olive oil, Cetapred ointment and clean water. A warning concerning the inadvertent use of the purple K type of fire extinguisher is required.
(ss) 815-Mercury. A mercury warning article shall be included in all manuals.
(tt) 816-Hot short. Include indentification and discussion of the hazardous by-products of the "hot short" cell such as smoke, toxic gases, spraying electrolyte, high temperature, etc.
(uu) 817-Sealing materials. Identify possible hazards present during preparation and application and cross reference to supporting information in chapter 7, section A.

30.8.6 Titles. Chapter and section titles shall be capital letters and shall have no additional text. Instructions and information shall be associated with a numbered article.

30.8.7 Cross references. Cross referencing of articles of related subjects shall be included to improve the effectiveness of the information presented.

30.9 List of illustrations. The illustrations provided for a specific manual shall be titled and arranged in the order as shown on example G. When a particular illustration is not appropriate the list shall contain that notation instead of the title on the appropriate line as follows:

(a) 5-Cell top connections. Intended to show connections for cooling water, etc. as applicable if not shown in cell assembly (1).
(b) 13-Charge voltage cut-off. Not required if this information is included in the master plan (12).
(c) 16-19-Hydrogen evolution. Effects of temperature and time on evolution to be provided as appropriate.
(d) 20-Effect of temperature. Relationship of initial temperature and charge acceptance to be described.
(e) 21-22-Effect of temperature and pressure. Pressure factor not applicable to all battery designs.

30.10 List of tables. The tables provided for a specific manual shall be titled and arranged in the order shown on example G. When a particular table is not appropriate the list shall contain that notation instead of the title on the appropriate line.

30.11 Appendices. Appendices shall be included in the manual to present instructions and information concerning the operation and use of auxiliary equipment such as the support van charging equipment for the Deep Submergence Rescue Vehicle (DSRV).

30.12 Format. Where appropriate, the format or arrangement of instructions or procedures obtained from another publication shall be used in the original form. Preparation of new text, where required, shall be in a manner similar to the basic organization of the service manual.

30.13 Identification. Appendices shall be identified individually as "A, B, C, etc." and so listed in the manual table of contents.

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NAVSEA ⓐ

SERVICE MANUAL

FOR

SILVER–ZINC BATTERY

ⓑ

FOR

ⓒ

Each transmittal of this document outside the agencies of the United States Government must have prior approval of NAVSEC.

DEPARTMENT OF THE NAVY
NAVAL SEA SYSTEMS COMMAND

EXAMPLE A. Cover.

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NAVSEA (a) _____

SERVICE MANUAL

OPERATING INSTRUCTIONS

FOR

SILVER-ZINC BATTERY

(b) ----------

FOR

(c) ----------

(d) ----------

CONTRACT N00024-XX-X-XXXX

Each transmittal of this document outside the agencies of the U.S. Government must have prior approval of NAVSEC.

DEPARTMENT OF THE NAVY

NAVAL SEA SYSTEMS COMMAND

PUBLICATION RECORD

ISSUE	DATE
Original	---------
CHG. # ----	---------
CHG. # ----	---------
CHG. # ----	---------
CHG. # ----	---------

EXAMPLE B. Title page.

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LIST OF EFFECTIVE PAGES

NOTE: On a changed page, the portion of the text affected by the latest change is indicated by a vertical line in the outer margin of the page. Changes to illustrations are indicated by miniature pointing hands.

Total number of pages in this manual is ____ consisting of the following:

Page No.	*Change No.
1-__ __	0

* Zero in this column indicates an original page.

EXAMPLE C. List of effective pages.

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CHANGE NO.	DATE	TITLE OR BRIEF DESCRIPTION	ENTERED BY

EXAMPLE D. Change record.

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APPROVAL AND PROCUREMENT RECORD PAGE

Approval Data For: NAVSEA XXXX-XX-XXX-XXXX

Title of Manual: Service Manual For Silver-Zinc Battery, (enter same data as title page, line 2)

Approval Authority: NAVSEC ltr 6157D:XXX:xxx N00024-XX-X-XXXX Ser XXXX-6157D dtd (enter date of ltr.)

CONTRACT NUMBER	SHIP APPLICABILITY	QUANTITY OF MANUALS	QUANTITY OF EQUIPMENT	INSTALLING ACTIVITY

REMARKS:

Date_____

CERTIFICATION:

(enter the appropriate certification paragraph from MIL-M-15071 in this space.)

MANUFACTURER'S SIGNATURE
MANUFACTURER'S NAME AND ADDRESS
FEDERAL CODE NUMBER

CHANGE NO.

EXAMPLE E. Approval and procurement record.

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VALIDATION PERFORMANCE

TITLE OF SERVICE MANUAL FOR SILVER-ZINC BATTERY

Contractor:

Contract No.

Chapter	Section	Article	Date validation completed	Check here if not validated
1	---	101 through 109		
2	A to E	201 through 268		
3	A to D	301 through 345		
4	A to D	401 through 442		
5	A to E	501 through 547		
6	A to D	601 through 632		
7	A to E	701 through 742		
8	A, B	801 through 817		

Names and authority of validating officer:

Signature of validating officer:

EXAMPLE F. Content assurance.

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TABLE OF CONTENTS

Chapter	Section and article	Page
	TITLE PAGE	1
	LIST OF EFFECTIVE PAGES	2
	CHANGE RECORD	3
	APPROVAL AND PROCUREMENT RECORD PAGE	4
	VALIDATION PERFORMANCE	5
	TABLE OF CONTENTS	6-
1	GENERAL INFORMATION	
	101 Abreviations and symbols	
	102 Identification	
	103 Description	
	104 Battery	
	105 Cell	
	106 Pressure compensation	
	107 Equipment supplied	
	108 Accessory equipment not supplied	
	109 Guarantee	
2	FUNCTIONAL DESCRIPTION	
	A ELECTROCHEMISTRY	
	201 General	
	202 Reactions	
	B CELL COMPONENTS	
	211 Cell case	
	212 Cell cover	
	213 Cell terminals	
	214 Electrolyte level indicator	
	215 Cooling panels	
	216 Firewalls	
	217 Flash arresters	
	218 Bubble breakers	
	219 Cell marking	
	220 Positive plate	
	221 Negative plate	
	222 Separator materials	
	223 Electrolyte	
	224 Oil	
	C BATTERY COMPONENTS	
	231 Battery box	
	232 Battery cover	
	233 Flushing valve	
	234 Temperature sensor	

EXAMPLE G. Table of contents.

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TABLE OF CONTENTS

Chapter	Section and article	Page
2 (con.)	235 Battery connector	
	236 Intercell connectors	
	D AUXILIARY EQUIPMENT	
	241 Cell voltage monitor	
	242 Voltmeters	
	243 Ammeters	
	244 Ampere-hour meter	
	245 Hydrogen detector	
	246 Ground detector	
	247 Thermometers	
	248 Discharge equipment	
	E MECHANISMS	
	261 Capacity	
	262 Voltage	
	263 Electrolyte level change	
	264 Heat generation	
	265 Gas evolution	
	266 Cell reversal	
	267 Capacity loss	
	268 Internal cell shorting	
3	PREPARATION AND INSTALLATION	
	A SHIPMENT	
	301 Dry shipment	
	302 Wet shipment	
	B STORAGE	
	311 Dry	
	312 Wet charged	
	313 Wet discharged	
	C PREPARATION FOR INSTALLATION	
	321 Safety precautions	
	322 Filling cells	
	323 Assembling battery	
	324 Formation	
	325 Encapsulation	
	326 Oil filling	

EXAMPLE G. Table of contents. (Continued)

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TABLE OF CONTENTS

Chapter	Section and article	Page
3 (con.)	D INSTALLATION	
	331 Parts list	
	332 Shipping list	
	333 Installation hardware list	
	334 Maintenance parts list	
	335 Special tools list	
	336 Installation	
	337 Wedging	
	338 Intercell connectors	
	339 Connect battery terminals	
	340 Connect voltage monitor leads	
	341 Connect water cooling system	
	342 Install level indicators	
	343 Install termperature sensors	
	344 Placing battery in service	
	345 Testing	
4	OPERATION	
	A GENERAL	
	401 Safety precautions	
	402 Auxiliary equipment	
	403 Jumpered cells	
	404 Parallel batteries	
	B CHARGING	
	411 General characteristics	
	412 State of charge	
	413 Normal charge	
	414 Overcharge	
	415 Partial charge	
	416 Float charge	
	417 Equalizing charge	
	418 Maintenance charge	
	419 Charge rate	
	420 Voltage limits	
	421 When to charge	
	422 Ventilation	
	423 Temperature limits	
	424 Charging procedures	

EXAMPLE G. Table of contents. (Continued)

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TABLE OF CONTENTS

Chapter	Section and article		Page
4 (con.)	C	DISCHARGING	
		431 General characteristics	
		432 Discharge rate	
		433 Voltage limits	
		434 Ventilation	
		435 Temperature limits	
		436 Capacity test discharge	
	D	BATTERY RECORDS	
		441 Battery record book	
		442 Battery monthly report	
5	MAINTENANCE		
	A	REQUIRED ACTION	
		501 General	
		502 Capacity maintenance	
		503 Preventive maintenance	
		504 Maintenance schedule	
	B	STANDARDS	
		511 Voltage	
		512 Capacity	
		513 Electrolyte level	
		514 Ground resistance	
		515 Meters	
		516 Ventilation system	
		517 Cell seals	
		518 Battery seals	
		519 General cleanliness	
	C	SHIPBOARD ACTION	
		521 Voltage	
		522 Capacity	
		523 Electrolyte level	
		524 Ground resistance	
		525 Meters	
		526 Ventilation	
		527 Cell seals	
		528 Battery seals	
		529 General cleanliness	

EXAMPLE G. Table of contents. (Continued)

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TABLE OF CONTENTS

Chapter	Section and article	Page
5 (con.)	D SUPPORT SHIP AND SHORE BASED ACTION	
	531 Voltage	
	532 Capacity	
	533 Electrolyte level	
	534 Ground resistance	
	535 Meters	
	536 Ventilation	
	537 Cell seals	
	538 Battery seals	
	539 General cleanliness	
	E RECOMMENDATIONS	
	541 General	
	542 Open circuit versus float	
	543 Booster charge	
	544 Temporary storage	
	545 Low temperature storage	
	546 Electrolyte level	
	547 Additional comments	
6	TROUBLE SHOOTING	
	A FAILURE MODES	
	601 Capacity loss	
	602 Internal shorting	
	B CAPACITY LOSS	
	611 Causes	
	612 Float charging	
	613 Equalizing charge	
	614 Maintenance charge	
	C INTERNAL SHORTING	
	621 Detection	
	622 Verification	
	623 Action	
	624 Trouble shooting diagram	
	D CELL REPLACEMENT	
	631 When to replace	
	632 How to replace	

EXAMPLE G. Table of contents. (Continued)

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TABLE OF CONTENTS

Chapter	Section and article	Page
7	REPAIR	
	A SEALS	
	701 General	
	702 Materials	
	703 Preparation	
	704 Application	
	B FLASH ARRESTER	
	711 Cleaning	
	712 Repair	
	713 Replacement	
	C REDUCING GROUNDS	
	721 General	
	722 Action	
	D REPLACING CELLS	
	731 Removing old cell	
	732 Preparing new cell	
	733 Installation	
	734 Battery reassembly	
	E RESTORING BATTERY CAPACITY	
	741 General	
	742 Procedure	
8	SAFETY PRECAUTIONS	
	A SAFETY EQUIPMENT	
	801 General	
	802 Fire extinguisher	
	803 Face shield or goggles	
	804 Neoprene sheeting	
	805 Insulated tools	
	806 Vaseline or petrolatum	
	807 First aid materials	
	808 Glass beaker or large test tube	
	809 Rags or towelling	

EXAMPLE G. Table of contents. (Continued)

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TABLE OF CONTENTS

Chapter	Section and article	Page
8 (con.)	B HAZARDS	
	811 Electrical	
	812 Hydrogen	
	813 Electrolyte	
	814 Seawater	
	815 Mercury	
	816 Hot short	
	817 Sealing and encapsulating materials	
	LIST OF ILLUSTRATIONS	
	LIST OF TABLES	
	USER ACTIVITY COMMENT SHEET	

EXAMPLE G. Table of contents. (Continued)

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LIST OF ILLUSTRATIONS

Number	Title	Page
1	Cell assembly	
2	Battery box	
3	Battery assembly	
4	Battery cell layout	
5	Cell top connections	
6	Battery cell interconnections	
7	Receptacle plate assembly	
8	Wiring diagram	
9	Level indicator and interpretation	
10	Temperature sensor assembly	
11	Electrolyte adjustment device	
12	Master plan, curves and data	
13	Charge voltage cut-off	
14	Typical charge curves	
15	Typical discharge curves	
16	Hydrogen evolution on stand	
17	Hydrogen evolution on float	
18	Hydrogen evolution on charge	
19	Hydrogen evolution on discharge	
20	Effect of temperature on charge	
21	Effect of temperature and pressure on discharge	
22	Effect of temperature and pressure on charge	

EXAMPLE G. Table of contents. (Continued)

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LIST OF TABLES

Number	Title	Page
1	Equipment supplied	
2	Other equipment required	
3	Electrolyte specification	
4	First aid materials	
5	Scheduled maintenance	
6	Installation kit	
7	Service kit repair parts	
8	Oil specification	

EXAMPLE G. Table of contents. (Continued)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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